Electrodynamics of Fractal Distributions of Charges and Fields

Fractional Continuous Models and Generalized Maxwell Equations

Based on V.E. Tarasov

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Motivation

- Classical electrodynamics assumes **smooth charge and current distributions** in $\mathbb{R}^3$
- Many physical systems are irregular or strongly inhomogeneous:
  - Turbulent plasmas
  - Porous or disordered media
  - Percolation clusters
  - Space plasmas and astrophysical environments
- These systems often exhibit **fractal geometry** with non-integer dimension

How should Maxwell's equations be modified when charges live on fractals?

What is a Fractal Distribution?

- A fractal set has a (typically non-integer) **Hausdorff dimension** D
- Charge does not fill 3D volume uniformly, but resides on a set of dimension D
- Scaling law for total charge inside a ball of radius R:

$$Q(R) \propto R^D, \qquad 0 < D \leq 3$$

- In ordinary media, $D = 3$ for a bulk, $D = 2$ for a surface, $D = 1$ for a line
- Here D can be non-integer and is defined operationally from the scaling

Standard Electrodynamics

- Charge density $\rho(\boldsymbol{r})$ defined on $\mathbb{R}^3$
- Volume, surface, and line elements:

$$dV = d^3x, \qquad d\boldsymbol{S}, \qquad d\boldsymbol{l}$$

- Total charge in region $W \subset \mathbb{R}^3$:

$$Q(W) = \int_W \rho(\boldsymbol{r})\, dV$$

- Current density $\boldsymbol{J}(\boldsymbol{r}, t)$ and total current through surface S:

$$I(S) = \int_S \boldsymbol{J} \cdot d\boldsymbol{S}$$

Standard Electrodynamics

Maxwell equations (differential form, in vacuum):

$$\nabla \cdot \boldsymbol{E} = \frac{\rho}{\varepsilon_0},$$
$$\nabla \cdot \boldsymbol{B} = 0,$$
$$\nabla \times \boldsymbol{E} = -\frac{\partial \boldsymbol{B}}{\partial t},$$
$$\nabla \times \boldsymbol{B} = \mu_0 \boldsymbol{J} + \mu_0 \varepsilon_0 \frac{\partial \boldsymbol{E}}{\partial t}.$$

Integral form (Gauss and Ampère as examples):

$$\oint_S \boldsymbol{E} \cdot d\boldsymbol{S} = \frac{Q(W)}{\varepsilon_0},$$
$$\oint_C \boldsymbol{B} \cdot d\boldsymbol{l} = \mu_0 I(S) + \mu_0 \varepsilon_0 \frac{d}{dt} \int_S \boldsymbol{E} \cdot d\boldsymbol{S}.$$

Space is smooth and three-dimensional at all scales.

Where to add Fractional Calculus?

- So far, no fractional *derivatives* appeared in Maxwell equations
- If we naively replaced $\partial \to \partial^\alpha$ we would break Lorentz invariance
- But we can use Integration of non-integer order
- This corresponds to *fractional measures* on $\mathbb{R}^3$

This approach follows Tarasov's

"fractional continuous model of fractal distributions."

Fractional Integrals and Fractal Measures

For a function $f(\boldsymbol{r})$, the Riesz fractional integral in $\mathbb{R}^3$ is

$$(I^D f)(\boldsymbol{r}) = C(D) \int_{\mathbb{R}^3} \frac{f(\boldsymbol{r}')}{|\boldsymbol{r} - \boldsymbol{r}'|^{3-D}} \, d^3 r', \qquad 0 < D < 3$$

- This integral has the same scaling as a D-dimensional volume
- It defines an **effective integration over a fractal**
- This is important since many fractals are singular sets! (Lebeasgue measure is 0)

Tarasov shows that

$$(I^D f)(\boldsymbol{r}) \iff \int_W f(\boldsymbol{r}) \, dV_D$$

Density of States as Fractional Kernel

Let's introduce the fractional volume element

$$dV_D = c_3(D, \boldsymbol{r})\, d^3x \qquad c_3(D, r) \propto r^{D-3}$$

It is exactly the kernel of a Riesz-type fractional integral

- Fractional integration replaces

$$\int d^3x \;\to\; \int r^{D-3} d^3x$$

- The order of integration D equals the fractal dimension
- Replace the fractal by an **effective continuous distribution** in $\mathbb{R}^3$

For a ball of radius R one has

$$\int_{|\boldsymbol{r}|\leq R} dV_D \propto R^D$$

Density of States $c_n(D, \boldsymbol{r})$

- $c_n(D, \boldsymbol{r})$: **density of states** in n-dimensional Euclidean space
- Tells us how densely "available positions" for charges are packed
- For $n = 3$, a typical choice (Riemann–Liouville type) is

$$c_3(D, r) = C(D)\, r^{D-3}$$

  where $r = |\boldsymbol{r}|$ and $C(D)$ is a dimension-dependent normalization factor
- For $D = 3$, $c_3(3, r) = 1$ and we recover the usual volume element
- Similar definitions exist for $c_2(d, \boldsymbol{r})$ on surfaces and $c_1(\gamma, \boldsymbol{r})$ on curves

Generalized Gauss and Stokes Theorems

- For a vector field $\boldsymbol{F}(\boldsymbol{r})$, Gauss theorem in this model is schematically

$$\oint_{\partial W} \boldsymbol{F} \cdot d\boldsymbol{S}_d = \int_W \nabla \cdot \left(c_2(d, \boldsymbol{r})\, \boldsymbol{F}(\boldsymbol{r})\right) c_3(D, \boldsymbol{r})\, d^3x$$

- Stokes theorem similarly becomes

$$\oint_{\partial S} \boldsymbol{F} \cdot d\boldsymbol{l}_\gamma = \int_S \left[\nabla \times \left(c_1(\gamma, \boldsymbol{r})\, \boldsymbol{F}(\boldsymbol{r})\right)\right] \cdot d\boldsymbol{S}_d$$

- The *form* of the theorems is preserved, but the divergence and curl are effectively weighted by c_n.

Charge and Current in Fractal Media

- Total fractal charge in region W:

$$Q_D(W,t) = \int_W \rho(\mathbf{r},t)\, c_3(D,\mathbf{r})\, d^3x$$

- Current density remains

$$\boldsymbol{J}(\mathbf{r},t) = \rho(\mathbf{r},t)\, \boldsymbol{u}(\mathbf{r},t)$$

- but the total current through a fractal surface S of dimension d is

$$I_d(S,t) = \int_S \boldsymbol{J}(\mathbf{r},t)\cdot d\boldsymbol{S}_d, \qquad d\boldsymbol{S}_d = c_2(d,\mathbf{r})\, d\boldsymbol{S}$$

Charge Conservation

Global charge conservation for a region W:

$$\frac{d}{dt} Q_D(W,t) = -I_d(\partial W, t).$$

In terms of densities:

$$\frac{d}{dt} \int_W \rho(\boldsymbol{r},t)\, c_3(D,\boldsymbol{r})\, d^3x = -\int_{\partial W} \boldsymbol{J}(\boldsymbol{r},t) \cdot d\boldsymbol{S}_d$$

Using generalized Gauss theorem and assuming W is fixed in time, this leads to a continuity equation:

$$c_3(D,\boldsymbol{r}) \frac{\partial \rho}{\partial t} + \nabla \cdot \left(c_2(d,\boldsymbol{r})\, \boldsymbol{J}(\boldsymbol{r},t) \right) = 0$$

For $D = 3$ and $d = 2$, $c_3 = c_2 = 1$ and we recover

$$\frac{\partial \rho}{\partial t} + \nabla \cdot \boldsymbol{J} = 0$$

Coulomb's Law for Fractal Sources

For a charge distribution $\rho(\boldsymbol{r}')$ with fractal dimension D, the electric field at $\boldsymbol{r}$ is

$$\boldsymbol{E}(\boldsymbol{r}) = \frac{1}{4\pi\varepsilon_0} \int_W \frac{\boldsymbol{r} - \boldsymbol{r}'}{|\boldsymbol{r} - \boldsymbol{r}'|^3}\, \rho(\boldsymbol{r}')\, c_3(D, \boldsymbol{r}')\, d^3x'$$

- The kernel is the same as in standard 3D electrostatics
- The *measure* $c_3(D, \boldsymbol{r}')\, d^3x'$ carries information about fractal geometry
- For homogeneous fractal distributions, one obtains field scalings like $E(R) \propto R^{D-3}$

Biot–Savart Law for Fractal Currents

For a steady current distribution $\boldsymbol{J}(\boldsymbol{r}')$ with fractal dimension D:

$$\boldsymbol{B}(\boldsymbol{r}) = \frac{\mu_0}{4\pi} \int_W \frac{\boldsymbol{J}(\boldsymbol{r}') \times (\boldsymbol{r} - \boldsymbol{r}')}{|\boldsymbol{r} - \boldsymbol{r}'|^3} \, c_3(D, \boldsymbol{r}') \, d^3x'$$

- Again, the kernel is unchanged
- Fractal integration modifies how strongly regions contribute as a function of distance
- In symmetric cases, one can get magnetic fields that decay with non-standard exponents or even stay constant

Gauss and Ampère Laws with Fractal Measures

Electric flux through a fractal surface $S = \partial W$:

$$\Phi_E(S, t) = \oint_S \boldsymbol{E}(\boldsymbol{r}, t) \cdot d\boldsymbol{S}_d = \frac{1}{\varepsilon_0} \int_W \rho(\boldsymbol{r}, t)\, dV_D$$

Ampère's law (without displacement current, for steady currents):

$$\oint_C \boldsymbol{B}(\boldsymbol{r}) \cdot d\boldsymbol{l}_\gamma = \mu_0 \int_S \boldsymbol{J}(\boldsymbol{r}) \cdot d\boldsymbol{S}_d$$

- For homogeneous fractal distributions, these lead to simple power-law fields
- Example: for suitable D and d, $E(R)$ or $B(R)$ can be constant with R

Maxwell Equations for Fractal Distributions (Integral Form)

Assuming charges and fields are defined on the fractal set:

$$
\begin{aligned}
&\oint_S \boldsymbol{E} \cdot d\boldsymbol{S}_d = \frac{1}{\varepsilon_0} \int_W \rho(\boldsymbol{r}, t)\, dV_D \\
&\oint_S \boldsymbol{B} \cdot d\boldsymbol{S}_d = 0 \\
&\oint_C \boldsymbol{E} \cdot d\boldsymbol{l}_\gamma = -\frac{d}{dt} \oint_S \boldsymbol{B} \cdot d\boldsymbol{S}_d \\
&\oint_C \boldsymbol{B} \cdot d\boldsymbol{l}_\gamma = \mu_0 \int_S \boldsymbol{J}(\boldsymbol{r}, t) \cdot d\boldsymbol{S}_d + \mu_0 \varepsilon_0 \frac{d}{dt} \oint_S \boldsymbol{E} \cdot d\boldsymbol{S}_d
\end{aligned}
$$

- Same structure as standard Maxwell equations, but with dV_D, $d\boldsymbol{S}_d$, $d\boldsymbol{l}_\gamma$.

Maxwell Equations for Fractal Distributions (Differential Form)

Using generalized Gauss and Stokes theorems, the integral equations lead to:

$$\nabla \cdot \left(c_2(d,\boldsymbol{r})\,\boldsymbol{E}(\boldsymbol{r},t)\right) = \frac{1}{\varepsilon_0} c_3(D,\boldsymbol{r})\,\rho(\boldsymbol{r},t)$$

$$\nabla \cdot \left(c_2(d,\boldsymbol{r})\,\boldsymbol{B}(\boldsymbol{r},t)\right) = 0$$

$$\nabla \times \left(c_1(\gamma,\boldsymbol{r})\,\boldsymbol{E}(\boldsymbol{r},t)\right) = -c_2(d,\boldsymbol{r})\,\frac{\partial \boldsymbol{B}}{\partial t}$$

$$\nabla \times \left(c_1(\gamma,\boldsymbol{r})\,\boldsymbol{B}(\boldsymbol{r},t)\right) = \mu_0 c_2(d,\boldsymbol{r})\,\boldsymbol{J}(\boldsymbol{r},t) + \mu_0\varepsilon_0 c_2(d,\boldsymbol{r})\,\frac{\partial \boldsymbol{E}}{\partial t}$$

- These are ordinary differential equations in space and time, but with position-dependent weights from geometry.

Apparent Magnetic Monopoles from Fractality

From

$$\nabla \cdot \big(c_2(d, \boldsymbol{r})\, \boldsymbol{B}\big) = 0$$

we can write

$$c_2(d, \boldsymbol{r})\, \nabla \cdot \boldsymbol{B} + \boldsymbol{B} \cdot \nabla c_2(d, \boldsymbol{r}) = 0$$

If c_2 varies in space (true fractal case), then

$$\nabla \cdot \boldsymbol{B} = -\frac{\boldsymbol{B} \cdot \nabla c_2}{c_2} \neq 0$$

in general.

- The field behaves *as if* there were effective magnetic charges
- These are not fundamental monopoles, but artifacts of the fractal geometry

Effective Medium Interpretation

Define effective fields

$$\boldsymbol{E}_{\text{eff}} = c_1(\gamma, \boldsymbol{r})\, \boldsymbol{E}, \qquad \boldsymbol{B}_{\text{eff}} = c_2(d, \boldsymbol{r})\, \boldsymbol{B}$$

Then we can rewrite Maxwell equations in a more standard form using

$$\boldsymbol{D} = \varepsilon_{\text{eff}}(\boldsymbol{r})\, \boldsymbol{E}_{\text{eff}}, \qquad \boldsymbol{H} = \frac{1}{\mu_{\text{eff}}(\boldsymbol{r})}\, \boldsymbol{B}_{\text{eff}}$$

- Fractal distribution behaves like a medium with spatially varying

  $$\varepsilon_{\text{eff}}(\boldsymbol{r}), \qquad \mu_{\text{eff}}(\boldsymbol{r}),$$

  determined by $c_n(D, \boldsymbol{r})$ and the fractal dimensions.
- Geometry (fractal) and material response become deeply intertwined.

Electric Multipole Expansion on Fractals

Scalar potential for a static fractal charge distribution:

$$\phi(\boldsymbol{R}) = \frac{1}{4\pi\varepsilon_0} \int_W \frac{\rho(\boldsymbol{r})\, dV_D}{|\boldsymbol{R} - \boldsymbol{r}|}$$

For $R = |\boldsymbol{R}|$ large compared to $|\boldsymbol{r}|$, expand

$$\frac{1}{|\boldsymbol{R} - \boldsymbol{r}|} = \sum_{n=0}^{\infty} \frac{r^n}{R^{n+1}} P_n(\cos\theta),$$

where θ is the angle between $\boldsymbol{R}$ and $\boldsymbol{r}$.

- Leads to usual monopole, dipole, quadrupole, … terms
- Coefficients are now integrals with respect to dV_D

Monopole and Dipole Moments

Monopole term (total charge):

$$\phi_0(\boldsymbol{R}) = \frac{1}{4\pi\varepsilon_0}\frac{Q_D(W)}{R}, \qquad Q_D(W) = \int_W \rho(\boldsymbol{r})\, dV_D$$

Dipole moment for fractal distribution:

$$\boldsymbol{p}^{(D)} = \int_W \boldsymbol{r}\,\rho(\boldsymbol{r})\, dV_D$$

Dipole contribution:

$$\phi_1(\boldsymbol{R}) = \frac{1}{4\pi\varepsilon_0}\frac{\boldsymbol{p}^{(D)}\cdot\boldsymbol{R}}{R^3}$$

- Structure identical to standard EM
- Numerical values change because dV_D redistributes weight in space
- In simple homogeneous examples, deviations from the $D = 3$ dipole are typically of order 10–20%.

Comparison with Standard Electrodynamics

Standard EM	**Fractal EM**
Space dimension 3	Effective dimension D (non-integer)
Ordinary measure d^3x	Fractional measure $dV_D = c_3(D, \boldsymbol{r})\, d^3x$
Uniform media	Geometrically inhomogeneous media
$\nabla \cdot \boldsymbol{B} = 0$	$\nabla \cdot (c_2 \boldsymbol{B}) = 0$ (effective monopoles)
Field scaling $E \sim 1/R^2$	$E \sim R^{D-3}$ (in simple symmetric cases)

Magnetohydrodynamics on Fractal Media

- Consider a plasma or conducting fluid whose *charged particles* are distributed on a fractal set of dimension D
- The electromagnetic field interacts with a **collective medium**, not isolated charges

Mass (or charge) conservation:

$$c_3(D, \boldsymbol{r}) \frac{\partial \rho}{\partial t} + \nabla \cdot \left(c_2(d, \boldsymbol{r}) \, \rho \, \boldsymbol{u} \right) = 0$$

Momentum balance (Euler equation):

$$c_3(D, \boldsymbol{r}) \, \rho \left(\frac{\partial \boldsymbol{u}}{\partial t} + (\boldsymbol{u} \cdot \nabla) \boldsymbol{u} \right) = -c_3(D, \boldsymbol{r}) \, \nabla p + \rho \, \boldsymbol{E} + \boldsymbol{J} \times \boldsymbol{B}$$

- Same structure as standard fluid dynamics
- Inertia and pressure are weighted by the fractal measure
- Lorentz force keeps its usual form

Induction Equation for Fractal Conducting Media

Using generalized Maxwell equations and Ohm's law

$$\boldsymbol{J} = \sigma\left(\boldsymbol{E} + \boldsymbol{u} \times \boldsymbol{B}\right),$$

where σ is conductivity, one obtains the **fractal induction equation**:

$$\frac{\partial}{\partial t}\left(c_2(d,\boldsymbol{r})\,\boldsymbol{B}\right) = \nabla \times \left[c_1(\gamma,\boldsymbol{r})\left(\boldsymbol{u} \times \boldsymbol{B}\right)\right] + \eta\,\nabla^2\left(c_2(d,\boldsymbol{r})\,\boldsymbol{B}\right)$$

where $\eta = 1/(\mu_0\sigma)$ is the magnetic diffusivity.

- Geometry modifies magnetic flux conservation
- Even ideal MHD ($\eta \to 0$) is affected by fractality
- Flux freezing becomes *scale-dependent*

Physical Interpretation and Limits

- Fractal geometry acts as an **effective medium**
- Transport, diffusion, and field amplification depend on D, d, γ
- Standard MHD is recovered for

$$D = 3, \quad d = 2, \quad \gamma = 1 \Rightarrow c_n = 1$$

- Deviations become important in:
  - Turbulent plasmas
  - Percolating current networks
  - Space and astrophysical plasmas

Thank you!